TRAINITY PROJECT 7

Impact Of Car Features

Project Description: Analysing the automotive industry and what understanding what factors drive consumer needs. Further analysing the data the understand how a manufacturer can increase consumer demand and sales in this day and age with alternative fuels starting to emerge yet gasoline being the primary in vehicles.

Tech-Stack Used: Microsoft Excel

Data Cleaning:

- Engine Fuel Type was missing in 3 rows, all for the vehicle Suzuki Verona (2004 model). The dataset was checked if any other rows contained this vehicle. It was found that this vehicle was related to a 'regular unleaded' fuel type. Therefore, the 3 rows were replaced by 'regular unleaded'.
- Several rows of data was missing in Engine HP, the same method as Engine Fuel Type was used, by tracking the Make, Model and Year of the car, if the HP was located, it was replaced by the respective value else if no similar data was found, then the rows were deleted to remove unnecessary blanks.
- Engine Cylinder column has a few blanks as well, most of the blanks were associated with EV's therefore, those were replaced by 0 as they do not work on cylinders. For gasoline vehicles, their peers were tracked to understand the relation between Make, Engine HP and Transmission to understand how many cylinders would be associated with the vehicle type.
- Transmission Type had a few rows where the type was mentioned as unknown, those were replaced by the types in their respective categories.
- Number Of Doors had missing values in 5 rows, they were replaced by number of doors found in the same category.
- The Market Category data had a lot of missing values,

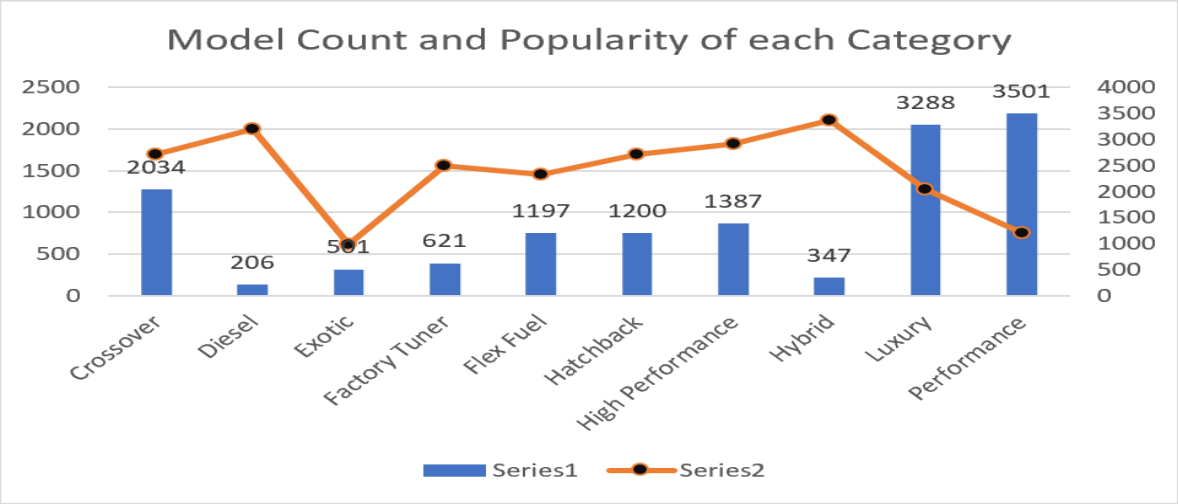
Outlier Detection: All the outliers here were detected by plotting box plots to figure out what values were outside the typical range and then they were analysed.

- There were a few outliers present in the Highway MPG column, one of them was for AUDI A6 which was replaced by the actual MPG mentioned for other AUDI A6 entries. The other outliers were present for electric vehicles for which the MPGe is known to be high and they do not follow the standard MPG format which is why they were left untouched.
- There were a few outliers present in the City MPG column as well, again the electric vehicles were exempted and the other outliers were dealt with either by deleting the row if no other data could be found or else replaced by data from similar models data within the dataset.
- MSRP data upon being looked at, indicates skewness but the MSRP column is subjective to the vehicle as all types of vehicles are included in this list varying all the way from 1990 where the prices could be significantly lower which is why nothing was done for this column.

Task 1.A: Create a pivot table that shows the number of car models in each market category and their corresponding popularity scores.

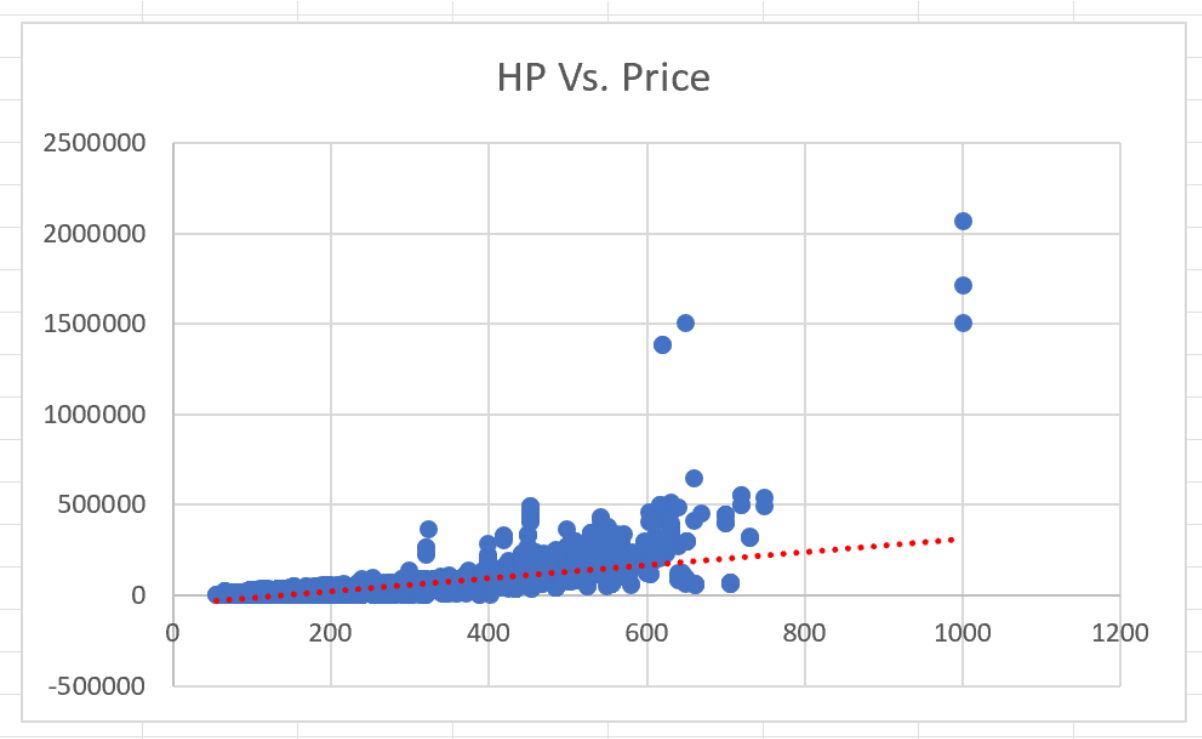
Task 1.B: Create a combo chart that visualizes the relationship between market category and popularity.

Market Category	Models	Popularity
Crossover	2034	1698.83
Diesel	206	2002.95
Exotic	501	617.00
Factory Tuner	621	1560.18
Flex Fuel	1197	1458.70
Hatchback	1200	1694.50
High Performance	1387	1821.44724
Hybrid	347	2105.57
Luxury	3288	1273.16
Performance	3501	751.94



- The most models amongst cars is in 'Performance' followed by 'Luxury' category. The least number of models is in 'Diesel' followed by 'Hybrid' category.
- The most popular category although based on the highest average rating is 'Hybrid' followed by'Diesel'. The least popular is 'Exotic' followed by 'Performance'.
- The reason for the the categories with the least number of total cars having the highest popularity might be due to lesser amount of reviews recorded. The exact comparison can probably be made when the same number of reviews are analysed for all categories.

Task 2: Create a scatter chart that plots engine power on the x-axis and price on the y-axis. Add a trendline to the chart to visualize the relationship between these variables.



- From the scatter plot, it can be seen that there is slight inscrease in the price of a car as the horse power increases. Some of the cars that have a horse power of 1000 can be seen to be priced much higher along with a few around the 600 horse power range as well.
- It can be understood that people like speed and as the vehicle gets faster, they are willing to pay more for it.

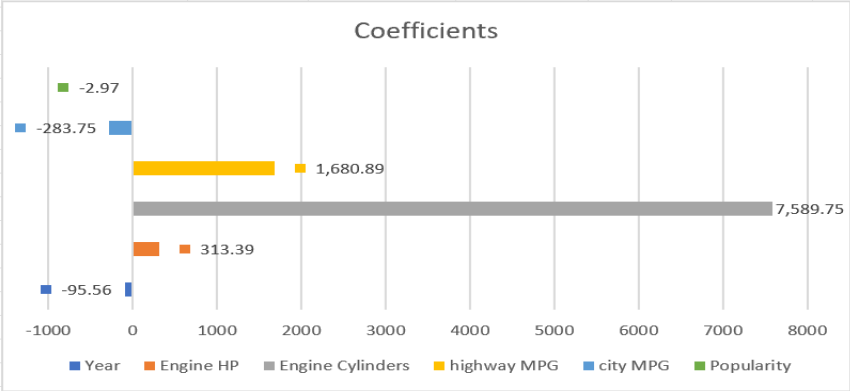
Task 3: Use regression analysis to identify the variables that have the strongest relationship with a car's price. Then create a bar chart that shows the coefficient values for each variable to visualize their relative importance.

Regression Statistics	
Multiple R	0.68049006
R Square	0.46306673
Adjusted R Square	0.46279598
Standard Error	44070.9524
Observations	11906

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	6	1.99314E+13	3.32191E+12	1710.34	0
Residual	11899	2.31108E+13	1942248850		
Total	11905	4.30423E+13			

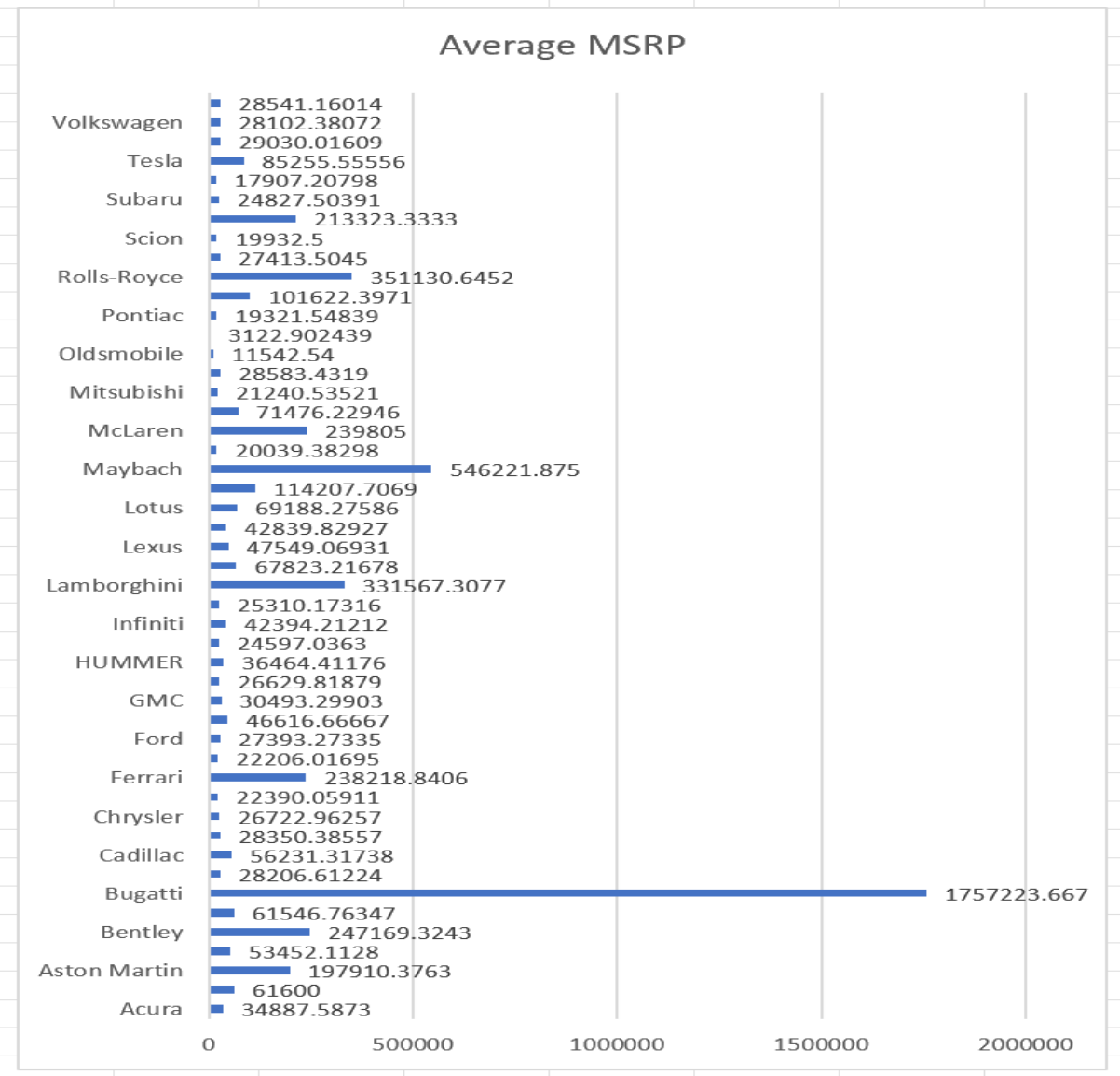
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	77397.9372	139050.2908	0.556618305	0.5778	-195163.3496	349959.224	-195163.3496	349959.2241
Year	-95.5648576	69.5700592	-1.373649221	0.16958	-231.9335394	40.8038242	-231.9335394	40.80382422
Engine HP	313.392622	7.171804299	43.69787692	0	299.3347134	327.45053	299.3347134	327.4505296
Engine Cylinders	7589.75048	472.7905503	16.05309259	2.2E-57	6663.003762	8516.4972	6663.003762	8516.4972
highway MPG	1680.88812	162.3729274	10.35202204	5.2E-25	1362.610658	1999.16559	1362.610658	1999.165588
city MPG	-283.75093	138.9822855	-2.041633787	0.04121	-556.1789153	-11.322944	-556.1789153	-11.32294447
Popularity	-2.97216669	0.282683546	-10.51411283	9.6E-26	-3.526272626	-2.4180608	-3.526272626	-2.418060762



- By performing a regression analysis of some of the attributes of a vehicle against the vehicle's price. A vehicles 'Engine Cylinders' have a big impact on the price indicating that as the number of cylinders in the engine increases, the price of the car significantly increases as well.
- The highway MPG along with Engine HP also indicate a positive impact on the vehicle's price showing an increase in price as they increase.
- Year, city MPG and Popularity do not have a positive impact on the cars price and show a negative impact.
- Therefore, features such as Engine Cylinders, highway MPG and Engine HP matter to the customers and lead to an increase in the vehicle's price.

Task 4.A: Create a pivot table that shows the average price of cars for each manufacturer.

Task 4.B: Create a bar chart or a horizontal stacked bar chart that visualizes the relationship between manufacturer and average price.

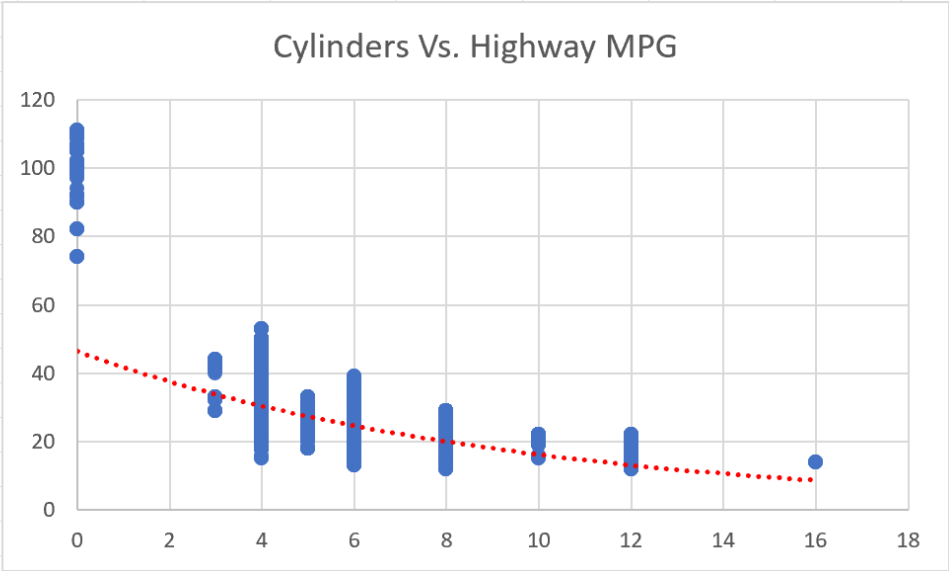


- Bugatti has the highest average price amongst all the producers, it is leading by a large margin. From our regression analysis we saw that Engine cylinders had a huge impact on a car's price, Bugatti is the only producer in the dataset offering a 16 cylinder engine which contributes a lot to a significantly higher price.
- Plymouth has the lowest average price, the reason why it has such a low average price is because they stopped manufacturing cars in 2000. The latest recording a car by Plymouth was in 2000 in the dataset when cars were a lot cheaper with no advanced technologies.

Task 5.A: Create a scatter plot with the number of cylinders on the x-axis and highway MPG on the y-axis. Then create a trendline on the scatter plot to visually estimate the slope of the relationship and assess its significance.

Task 5.B: Calculate the correlation coefficient between the number of cylinders and highway MPG to quantify the strength and direction of the relationship.

Correlation
-0.658486



- A negative correlation can be noted here, between Cylinders and Highway MPG.
- There is a downward trend in the scatter plot indicating that with an increase in the number of cylinders, the Highway MPG decreases as engines with higher cylinders require more gasoline to function because they produce a lot more horsepower and torque.
- The initial part of the curve, for 0 horsepower can be seen at a very high point, this is because the cars 0 horsepower are electric vehicles in this dataset which have a very high MPGe.

DASHBOARD

Task 1: How does the distribution of car prices vary by brand and body?

Sum of MSRP	Body Style																
Brand	**2dr Hatchback**	**2dr SUV**	**4dr Hatchback**	**4dr SUV**	**Cargo Minivan**	**Cargo Van**	**Convertible**	**Convertible SUV**	**Coupe**	**Crew Cab Pickup**	**Extended Cab Pickup**	**Passenger Minivan**	**Passenger Van**	**Regular Cab Pickup**	**Sedan**	**Wagon**	**Grand Total**
Acura	480917		357440	2663505					793748						4294702	201360	8791672
Alfa Romeo							129800		178200								308000
Aston Martin							7321655		9635275						1448735		18405665
Audi	4000			2674900			3291405		3556290						7158348	847350	17532293
Bentley							6012870		6356760						5920900		18290530
BMW	80097		1144950	3160950			4502671		3419051						7989300	259600	20556619
Bugatti									5271671								5271671
Buick				2141770			179325		18534			330065			2850590	8212	5528496
Cadillac				7182555			985607		2953574	599150					9418847	1184100	22323833
Chevrolet	8000	213310	1287260	6569568	420150	78688	2953245	106300	3504525	5927617	3117951	1178515	607670	2260032	3303977	300675	31837483
Chrysler	98805			250545			630105		114510			922295			2479859	501075	4997194
Dodge	48000	44000	18000	2572405	60520	338497	12000		3264627	2235775	864172	557425	70708	719408	2417585	793055	14016177
Ferrari							4723811		11713289								16437100
FIAT	325315			369305			327965									287570	1310155
Ford	36000	479873	480155	4482771	702400	566351	730007		1398144	3812353	2285584	1411605	2431898	1299240	2299348	1635565	24051294
Genesis															139850		139850
GMC		144319		6641919	142750	468085				4062482	2183866	150630	603670	1306328			15704049
Honda	413200		2015270	3953209			252135		1588705	787720		553185			2340105		11903529
HUMMER				377490						242405							619895
Hyundai	1038050		528880	2128890					724070			133075			2899937		7452902
Infiniti				4340200			980050		2175750						6494090		13990090
Kia			406960	2049645					142630			494650			1980360	772405	5846650
Lamborghini							7064450		10177050								17241500
Land Rover		476394		9076595				145731									9698720
Lexus			94700	3152974			472065		1016472						4837596	31105	9604912
Lincoln				3422570					25342	453260					2854855	269705	7025732
Lotus							413260		1593200								2006460
Maserati				155000			2342963		1972284						2153800		6624047
Maybach							2762750								5976800		8739550
Mazda	22000	24000	853180	3222525			870505		543879		580033	443130		265486	1618571	33350	8476659
McLaren							280225		918800								1199025
Mercedes-Benz			122800	4974610	28950		5753964		6473107			32500			7080243	764935	25231109
Mitsubishi	394868		407835	2066505	2000		209893			240210	134360	2000		8000	1058563		4524234
Nissan	14683		1347320	4149630	128620		1406552	131075	2943632	2422300	1026379	413320		21914	1769130	175000	15949555
Oldsmobile				238150			2000		286015			492055			691161	22000	1731381
Plymouth	42000		16000				85631		14000			33688			46759	18000	256078
Pontiac	163505		162975	401550			473481		667715			541192			1160535	22855	3593808
Porsche	28827			1815200			4504586		4758533						2713500		13820646
Rolls-Royce							2141365		2204675						6539010		10885050
Saab	14000		36586	541905			632628								1066500	751280	3042899
Scion	366325		282470						330210						32500	184445	1195950
Spyker							219990		419980								639970
Subaru	12000		678060	3020230					356476	365975					1913100	10000	6355841
Suzuki	46496	14000	584387	2362141				122194		304131	259659				1906715	685707	6285430
Tesla															1534600		1534600
Toyota	473750		1397750	5106450			386668		811995	3893760	3558504	1956518		373446	2459596	1237955	21656392
Volkswagen	4171275		3355760	2084955			3612631		8000			1038130			6760050	1704025	22734826
Volvo	157550			3219000			121600		6000						2086945	2428971	8020066
Grand Total	**8439663**	**1395896**	**15578738**	**100569617**	**1485390**	**1451621**	**66789858**	**505300**	**92336718**	**25347138**	**14010508**	**10683978**	**3713946**	**6253854**	**119697062**	**15130300**	**483389587**

Task 2: Which car brands have the highest and lowest average MSRPs, and how does this vary by body style?

Average of MSRP	Column Labels																
Row Labels	**2dr Hatchback**	**2dr SUV**	**4dr Hatchback**	**4dr SUV**	**Cargo Minivan**	**Cargo Van**	**Convertible**	**Convertible SUV**	**Coupe**	**Crew Cab Pickup**	**Extended Cab Pickup**	**Passenger Minivan**	**Passenger Van**	**Regular Cab Pickup**	**Sedan**	**Wagon**	**Grand Total**
Acura	17175.61		51062.86	42959.76					39687.40						33292.26	33560.00	34887.59
Alfa Romeo							64900.00		59400.00								61600.00
Aston Martin							203379.31		192705.50						206962.14		197910.38
Audi	2000.00			48634.55			70029.89		93586.58						44461.79	33894.00	53452.11
Bentley							250536.25		254270.40						236836.00		247169.32
BMW	26699.00		54521.43	58536.11			63417.90		51803.80						70701.77	43266.67	61546.76
Bugatti									1757223.67								1757223.67
Buick				33996.35			25617.86		2059.33			30005.91			27946.96	2053.00	28206.61
Cadillac				72551.06			70400.50		45439.60	66572.22					50912.69	47364.00	56231.32
Chevrolet	2000.00	8887.92	18930.29	32046.67	20007.14	7153.45	62835.00	17716.67	38939.17	39255.74	24170.16	24552.40	24306.80	19824.84	20521.60	15825.00	28350.39
Chrysler	32935.00			35792.14			24234.81		19085.00			29751.45			26103.78	26372.37	26722.96
Dodge	2000.00	2000.00	2000.00	30992.83	20173.33	12536.93	2000.00		45980.66	31052.43	13938.26	25337.50	14141.60	9342.96	21780.05	24782.97	22390.06
Ferrari							214718.68		249218.91								238218.84
FIAT	19136.18			24620.33			23426.07									22120.77	22206.02
Ford	2000.00	13710.66	18467.50	41507.14	21284.85	17698.47	34762.24		34101.07	41438.62	23808.17	23526.75	32425.31	17797.81	21290.26	27259.42	27393.27
Genesis															46616.67		46616.67
GMC		5550.73		36695.69	23791.67	18723.40				39062.33	26632.51	25105.00	26246.52	21069.81			30493.30
Honda	17216.67		25836.79	28855.54			36019.29		21763.08	34248.70		36879.00			26001.17		26629.82
HUMMER				37749.00						34629.29							36464.41
Hyundai	18536.61		17629.33	30412.71					20687.71			26615.00			27102.21		24597.04
Infiniti				45686.32			46669.05		40291.67						40588.06		42394.21
Kia			19379.05	31533.00					20375.71			32976.67			23298.35	20326.45	25310.17
Lamborghini							336402.38		328291.94								331567.31
Land Rover		39699.50		70910.90				48577.00									67823.22
Lexus			31566.67	45042.49			52451.67		50823.60						48864.61	31105.00	47549.07
Lincoln				50331.91					2111.83	41205.45					42609.78	44950.83	42839.83
Lotus							51657.50		75866.67								69188.28
Maserati				77500.00			130164.61		116016.71						102561.90		114207.71
Maybach							1381375.00								426914.29		546221.88
Mazda	2000.00	2000.00	20809.27	27080.04			28080.81		20143.67		11600.66	23322.63		9154.69	19738.67	16675.00	20039.38
McLaren							280225.00		229700.00								239805.00
Mercedes-Benz			40933.33	68145.34	28950.00		104617.53		109713.68			32500.00			49168.35	44996.18	71476.23
Mitsubishi	13162.27		13155.97	26158.29	2000.00		29984.71			26690.00	19194.29	2000.00		2000.00	24058.25		21240.54
Nissan	2097.57		24059.29	34294.46	21436.67		39070.89	43691.67	34228.28	32733.78	20527.58	22962.22		2191.40	21841.11	17500.00	28583.43
Oldsmobile				34021.43			2000.00		9226.29			32803.67			8131.31	2000.00	11542.54
Plymouth	2000.00		2000.00				28543.67		2000.00			2105.50			2597.72	2000.00	3122.90
Pontiac	18167.22		18108.33	25096.88			22546.71		15528.26			20815.08			20009.22	5713.75	19321.55
Porsche	5765.40			82509.09			115502.21		99136.10						123340.91		101622.40
Rolls-Royce							428273.00		367445.83						326950.50		351130.65
Saab	2000.00		2032.56	41685.00			28755.82								36775.86	34149.09	27413.50
Scion	20351.39		15692.78						27517.50						16250.00	18444.50	19932.50
Spyker							219990.00		209990.00								213323.33
Subaru	2000.00		21189.38	29322.62					15498.96	24398.33					26570.83	2000.00	24827.50
Suzuki	6642.29	2000.00	16696.77	21090.54				7187.88		27648.27	21638.25				18159.19	15237.93	17907.21
Tesla															85255.56		85255.56
Toyota	18950.00		22186.51	40851.60			25777.87		15615.29	37803.50	26359.29	29201.76		16236.78	24844.40	31742.44	29030.02
Volkswagen	24251.60		27964.67	41699.10			27789.47		2000.00			25320.24			29911.73	25818.56	28102.38
Volvo	26258.33			45338.03			40533.33		2000.00						20869.45	24785.42	28541.16
Grand Total	**16778.65**	**10115.19**	**22351.13**	**40421.87**	**20920.99**	**15280.22**	**84224.28**	**17424.14**	**76248.32**	**37220.47**	**22488.78**	**25621.05**	**29015.20**	**15953.71**	**39270.69**	**25557.94**	**40600.50**

Task 3: How do the different feature such as transmission type affect the MSRP, and how does this vary by body style?

Body Style	AUTOMATED_MANUAL	AUTOMATIC	DIRECT_DRIVE	MANUAL	Grand Total
2dr Hatchback	27180.96	20926.46		13316.32	16778.65
2dr SUV		17261.52		6303.81	10115.19
4dr Hatchback	29249.07	23833.68	32902.50	17594.41	22351.13
4dr SUV	40451.15	41535.61	49800.00	15426.46	40421.87
Cargo Minivan		20920.99			20920.99
Cargo Van		15280.22			15280.22
Convertible	121256.64	90374.37		62193.30	84224.28
Convertible SUV		38925.50		9233.14	17424.14
Coupe	245977.43	63249.80		50400.92	76248.32
Crew Cab Pickup		37744.07		28360.53	37220.47
Extended Cab Pickup		30559.34		10918.87	22488.78
Passenger Minivan		26412.68		4405.33	25621.05
Passenger Van		29015.20			29015.20
Regular Cab Pickup		28536.82		7366.95	15953.71
Sedan	47498.71	43769.12	79512.25	17092.71	39270.69
Wagon	31985.28	27613.19	34250.00	17844.14	25557.94

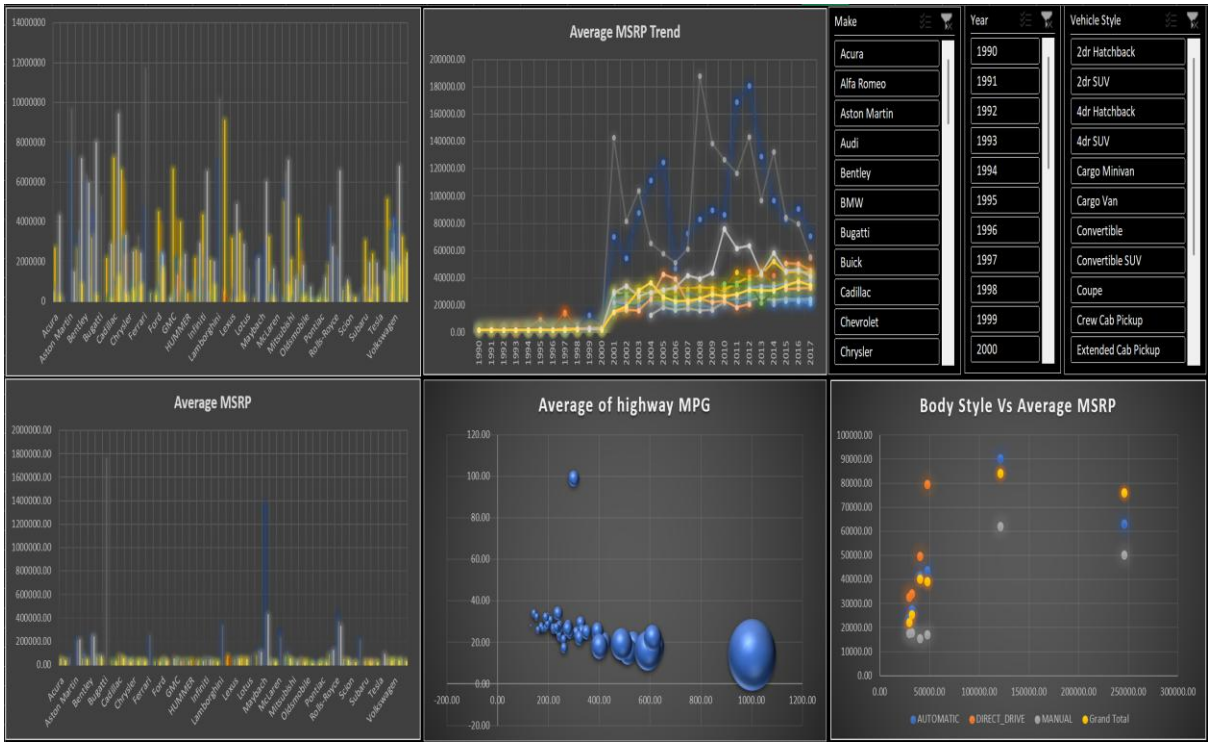
Task 4: How does the fuel efficiency of cars vary across different body styles and model years?

Average of MSRP	Column Labels																
Row Labels	2dr Hatchback	2dr SUV	4dr Hatchback	4dr SUV	Cargo Minivan	Cargo Van	Convertible	Convertible SUV	Coupe	Crew Cab Pickup	Extended Cab Pickup	Passenger Minivan	Passenger Van	Regular Cab Pickup	Sedan	Wagon	Grand Total
1990	2133.40	2000.00	2000.00		2000.00		2270.75		2023.07		2000.00	2000.00		2000.00	2011.77	2000.00	2020.20
1991	2069.80	2000.00		2000.00			2660.63		2032.74		2000.00	2000.00		2174.62	2035.12	2014.21	2080.66
1992	2000.00	2229.12	2000.00	2000.00			2395.75		2017.71		2009.30			2000.00	2093.60	2018.80	2077.74
1993	2143.13	2257.37	2000.00	2000.00			2842.00	2000.00	2380.22		2096.00			2010.65	2115.28	2041.85	2187.33
1994	2305.10	3930.36	2000.00	2000.00	2000.00	2000.00	2931.25	2000.00	2482.25		2234.71	2000.00	2000.00	2066.00	2009.00	2000.00	2312.03
1995	3057.14	9668.75	2000.00		2000.00	2000.00	3169.25	2000.00	2920.81		2408.55	2023.50	2000.00	2358.60	2072.77	2000.00	2606.78
1996	2000.00	2603.60	2000.00	2333.40	2000.00	2000.00	2451.00	2000.00	3484.36		2634.38	2109.22	2000.00	2123.40	2106.55	2023.56	2497.37
1997	2804.89	14556.33	2041.40	2422.20	2000.00	2000.00	2410.71	13643.67	3919.38		2968.86	2215.33	2065.67	2652.14	2126.41	2018.20	3105.21
1998	3816.40	2000.00	2086.00	2235.00		2100.20	3154.67	2000.00	5635.13		3245.53	2019.00	2242.00	2918.88	2343.36	2200.00	2975.60
1999	3562.44	2221.00		2760.80		2158.33	12649.00		3010.11		3580.65	2119.50		3245.50	2510.00		3096.16
2000	4113.58	2442.50		3071.00		2282.80	7297.71		3640.00		3382.25	2226.33	2329.00	3189.83	2575.96	2000.00	3166.67
2001	19820.07	22328.33		30380.45	20220.00	20166.00	70327.81		142830.71		21868.57	26468.00	22352.50	14650.71	28884.19	15249.00	41500.67
2002	27942.50	24316.88		27086.82	20585.00	20746.00	54440.57	17136.57	81695.70	33579.67	20175.00	27844.69	21345.00	16506.53	34022.92	18938.78	33744.16
2003	20047.50	24473.33		29548.26	22752.50	21506.00	87866.31	17307.60	104197.70	24520.00	19952.50	28056.76		16350.00	26568.63	30686.25	39774.06
2004	21883.57	23673.75	12880.00	28604.71	23676.00		111790.70		65457.63	24122.50	25797.50	28255.25		24473.46	29564.96	36479.00	36107.34
2005	21136.67	24146.67	18580.00	30442.52	23050.00		125030.64		57836.13	24485.00		26964.63		42555.00	31166.25	26334.94	36746.97
2006	20334.17		16784.50	33327.51	20310.00		47221.43		51448.56	31011.54		25867.05		39240.00	32697.20	22890.00	32056.47
2007	20549.62		17786.92	32046.05	20786.67		72603.12		61480.10	31167.17	28230.93	25744.17		21758.46	41617.62	23410.73	34974.19
2008	20536.59		16364.50	33384.01	23080.00		83071.27		187898.97	30738.15	25073.25	24971.43		24830.56	39284.20	24637.00	54138.92
2009	17845.00		16839.00	32228.69			89635.88		138652.03	31219.46	23877.29			22198.93	43591.20	28048.33	49838.56
2010	21753.46		22228.44	31299.60			86544.90		126829.39	32190.00	23280.00	35810.00		23181.36	75979.44	26476.09	50642.50
2011	24097.67		24273.34	36084.25			169168.72		116819.18	44218.15	22633.80	35322.86		18640.00	61821.24	28092.00	57548.18
2012	26614.13		23432.11	40926.40		28850.83	181047.23	44540.00	143364.03	39312.93	24076.63	35581.43	32652.50	21112.50	63441.06	31476.32	59516.24
2013	27157.11		25430.70	42414.12		29777.50	128920.55	44540.00	96704.14	44734.77		21855.00	33502.50		43313.38	30860.83	48731.58
2014	20713.52		23845.76	52291.31		30964.29	96754.06	41995.00	132372.97	35716.00	33116.00	30104.17	33797.81		58236.20	31007.71	63271.60
2015	24464.20	50866.67	24778.02	44130.64	23053.33	31901.25	83300.19		84292.64	37740.78	32040.55	32843.55	36105.36	31620.00	45047.80	33966.72	46809.51
2016	22968.52	50225.00	24450.93	44622.73	23734.62	32941.25	90531.15		79723.82	38900.43	32997.81	33748.55	36806.79	32416.32	45985.57	37497.13	47236.28
2017	21463.67	45700.00	24808.01	43566.80	23565.00		70599.47	54400.00	55682.82	40890.06	34556.62	33900.77	38567.50	32821.62	41206.15	34444.05	42205.62
Grand Total	**16778.65**	**10115.19**	**22351.13**	**40421.87**	**20920.99**	**15280.22**	**84224.28**	**17424.14**	**76248.32**	**37220.47**	**22488.78**	**25621.05**	**29015.20**	**15953.71**	**39270.69**	**25557.94**	**40600.50**

Task 5: How does the car's horsepower, MPG, and price vary across different Brands?

Make	Average of Engine HP	Average of highway MPG	Average of MSRP
Acura	244.80	28.11	34887.59
Alfa Romeo	237.00	34.00	61600.00
Aston Martin	484.32	18.89	197910.38
Audi	277.70	27.85	53452.11
Bentley	533.85	18.91	247169.32
BMW	326.91	29.25	61546.76
Bugatti	1001.00	14.00	1757223.67
Buick	219.24	26.95	28206.61
Cadillac	332.31	25.24	56231.32
Chevrolet	247.28	25.82	28350.39
Chrysler	229.14	26.37	26722.96
Dodge	244.42	22.35	22390.06
Ferrari	511.96	15.72	238218.84
FIAT	143.56	33.92	22206.02
Ford	242.47	23.75	27393.27
Genesis	347.33	25.33	46616.67
GMC	259.84	21.40	30493.30
Honda	195.75	32.25	26629.82
HUMMER	261.24	17.29	36464.41
Hyundai	201.92	30.39	24597.04
Infiniti	310.07	24.78	42394.21
Kia	208.84	30.65	25310.17
Lamborghini	614.08	18.02	331567.31
Land Rover	322.10	22.13	67823.22
Lexus	277.42	25.88	47549.07
Lincoln	284.09	24.49	42839.83
Lotus	275.97	26.55	69188.28
Maserati	420.79	20.29	114207.71
Maybach	590.50	16.00	546221.88
Mazda	171.99	27.85	20039.38
McLaren	610.40	22.20	239805.00
Mercedes-Benz	349.87	24.83	71476.23
Mitsubishi	172.92	27.54	21240.54
Nissan	241.00	27.80	28583.43
Oldsmobile	177.47	26.23	11542.54
Plymouth	131.56	27.96	3122.90
Pontiac	190.30	27.07	19321.55
Porsche	392.79	25.37	101622.40
Rolls-Royce	487.55	19.13	351130.65
Saab	220.52	26.35	27413.50
Scion	154.43	32.30	19932.50
Spyker	400.00	18.00	213323.33
Subaru	197.31	28.68	24827.50
Suzuki	160.29	26.03	17907.21
Tesla	300.00	98.94	85255.56
Toyota	235.93	26.45	29030.02
Volkswagen	189.76	32.13	28102.38
Volvo	230.97	27.20	28541.16

Task: Interactive Dashboard



- The final dashboard looks like this, with slicers attached to each of the charts, multi-select is allowed for the slicers, therefore you can filter out the headings for which you want to view the data.

Result: Through this project, I learnt about the complexities of data cleaning and anomaly handling. Understanding how to deal with these in context of the data to ensure analysis of the highest accuracy. I understood how to indentify skewed data and how to handle it. I understood the concept of pivot tables as well, how to manipulate data using a pivot table and create meaningful pivot charts. Alongside these, I also learned about slicers and the concept of story-telling and interactive dashboards.

Excel Link:

https://docs.google.com/spreadsheets/d/1Cm9QFoXMmqdJChxCpSBot3TwUveoS2eH/edit?usp=sharing&ouid=100878477496382804997&rtpof=true&sd=true

https://docs.google.com/spreadsheets/d/1CCs3UvnPL2kMeqvUBPM_OBOxzrxZTBW0/edit?usp=sharing&ouid=100878477496382804997&rtpof=true&sd=true